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27/09/2026

PRE - RELEASE W01 - W04 Diagnostic

④ ML Introduction:

1) Overfitting vs. Underfitting:

- + Overfitting $\left\{ \begin{array}{l} \text{Model too complex} \rightarrow \text{memorizes noises} \\ \text{Training Error: Low} \\ \text{Test Error: High} \end{array} \right.$
- + Underfitting $\left\{ \begin{array}{l} \text{Model too simple} \rightarrow \text{fails to capture objects} \\ \text{Training Error: High} \\ \text{Test Error: High} \end{array} \right.$
- + Good fit $\left\{ \begin{array}{l} \text{Model captures objects successfully, no memorize noise.} \\ \text{Balanced Training error and Test error.} \end{array} \right.$

2) Classify Metric: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$; $\text{Precision} = \frac{TP}{TP + FP}$; $\text{Recall} = \frac{TP}{TP + FN}$
 $\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

~~F1-score~~ where: + FP: a bounding box no matching ground-truth object exists in that location

+ FN: miss or fail to generate a valid bounding box although the real object exists $\rightarrow$ infinite. the number of dataset

- 3) Use:
- Accuracy: use when data is balanced.
 - Precision: need to limit ~~FP~~ / minimize FP.
 - Recall: need to minimize FN
 - F1-score: when data is seriously imbalanced.

3) Bias-Variance Decomposition:

$$E[(y - \hat{f}(x))^2] = \underbrace{\left(E[\hat{f}(x)] - f(x) \right)^2}_{\text{Bias}^2} + \underbrace{E\left[\left(\hat{f}(x) - E[\hat{f}(x)] \right)^2 \right]}_{\text{Variance}} + \underbrace{\epsilon^2}_{\text{Noise}}$$

If Bias decrease $\downarrow \rightarrow$ Variance increase $\uparrow$

* Decision Tree:

1) Classification Tree vs. Regression Tree:

	Classification Tree	Regression Tree
Traversal	From root to leaf	From root to leaf
leaf stores	Class Label / Class distribution (E.g.: Play = Yes/No)	Numeric Value (E.g.: mean of targets)
Prediction	Majority class at the leaf / Confidence from class frequencies	Mean/Median of training target.

2) Entropy: Measure the uncertainty (impurity) of dataset S .

$$H(S) = - \sum_{i=1}^K (p_i \cdot \log_2 p_i) \quad \leftarrow \begin{array}{l} p_i: \text{proportion of samples in class } i \\ \text{High entropy} \rightarrow \text{more mixed class} \\ \text{Low entropy} \rightarrow \text{pure class} \end{array}$$

3) IG: Measure the reduction in entropy after splitting by attribute A .

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

→ Attribute has the highest IG → Chosen for splitting.

4) Gini: Use in CART.

$$Gini(S) = 1 - \sum_{i=1}^K p_i^2 \quad \rightarrow \text{If all samples} \in 1 \text{ class (pure)} \rightarrow Gini(S) = 0$$

5) ID3: use Entropy and IG: - Greedy: IG highest → chosen

- Recursive splitting until: → All samples $\in 1$ class.

→ No attributes remain

6) C4.5: use Gain Ratio (GR) and have pruning → avoid overfitting

$$GR(S, A) = \frac{IG(S, A)}{\text{SplitInfo}(S, A)}; \quad \text{SplitInfo}(S, A) = - \sum_{v \in \text{Values}(A)} \left(\frac{|S_v|}{|S|} \cdot \log_2 \frac{|S_v|}{|S|} \right)$$

Step 1: Determine candidate thresholds.

Step 2: Split dataset

Step 3: Calculate IG.

Step 4: Recursive 1 → 3.

Step 5: Choose threshold T^* that is optimal (highest IG).

Pre-pruning: Node has low support: $|S| < N_{min}$; majority proportion $\geq \tau$.

Post-pruning: Traverse bottom-up

$$\begin{aligned} \text{leaf error (if pruned to single leaf): } err_{leaf}(t) &= N - \max_c n_{c,t} \\ \text{subtree error: } err_{subtree}(t) &= \sum_{l \in \text{leaves}(t)} (N_l - \max_c n_{c,l})^c \end{aligned}$$

total sample numbers of classes
↑
N { $n_{c,t}$ }

→ prune if $err_{leaf}(t) \leq err_{subtree}(t)$